Eric Moore, Noah Turock, Rachel Mohamed, Anthony Chapman

Malda

Abstract

Malda is a 2D Puzzle-Platforming game that sees players navigate platforms, fight enemies, and overcome obstacles to reach their goal. Upon completing a level, players will be awarded a new item to help them continue their journey. Some levels require items from previous levels to be completed. Levels may include optional portions that require items from later in the game to discover. Malda will feature a minimalistic pixel art style.

Malda players will need to be creative with their use of items, sometimes in combination with each other, in order to succeed. As players make it further in the game, they will find their technical platforming skills tested.

Description

Description

The core experience while playing Malda can best be described as a story-based, creative puzzle-solving platformer. Rather than relying on fast-paced technical gameplay, Malda will feature satisfying casual platforming. The challenge will come from the player's ability to creatively utilize their items and capabilities to overcome the challenges created by their environment. As players accomplish more challenges, they will acquire more items and/or abilities, adding to the complexity of the problems they can solve in future levels.

Items that players can unlock may be a boomerang, climbing boots, a bounce pad, a water gun, and a lasso. The boomerang is an early unlock for players to combat enemies more easily, as well as destroy various objects in the game. It can be used to hit buttons or levers and generally return to the player, as long as nothing sturdy is in the way. Climbing boots are another early unlock that can be used to augment platforming and enable players to briefly grab onto a wall before jumping off or to climb steep or slippery surfaces. The last early unlock will be a bounce pad. The bounce pad will allow players to jump off higher platforms and bounce back up to their original height while trying to traverse challenging sections. The water gun and lasso are unlocked later, requiring one or more of the previous items to unlock. The water gun will be used to push enemies and objects without destroying them, as well as put out fires located

throughout levels. The Lasso will be used to pull enemies towards the player or grab a ledge and swing around, greatly increasing mobility.

A high priority of Malda is how platforming feels for the player, so items and basic movement abilities will be carefully tuned to maximize player satisfaction while navigating the game. The goal is to slowly increase the difficulty as players become more comfortable with their new abilities.

Story

Malda, a being of the cosmos, has been sent to repair the connections between worlds, collectively known as the Tree of Life. There have been reports of 6 different worlds attached to these branches and roots, plunging into chaos as trickster beings have managed to mess with them. As you reach the center of the tree, you find it requires porthole keys to unlock your way to the next. Each key can be found inside the world you enter for each level.

How It's Entertaining

The Player will solve engaging puzzles using the various items and movement tech at their disposal. Each puzzle may have multiple solutions, utilising the various items in different ways, but it will be up to the player to determine how they will solve their way forward. Through practicing the movement of the game and mastering it, players will feel a sense of accomplishment by completing complex movement maneuvers using their items and their environment. New areas offer new ways to approach the movement of the game, and it will be up to the player to master the different ways to move. As players collect more and more items from across the worlds, players feel more accomplished and have new ideas to approach obstacles new and old.

Our story will use various role-playing elements to tell an engaging story centering on its tree of life theme. Different NPCs from different worlds will show how each world was thrown into chaos. Additionally, some NPCs may join you in the hub world when their world is saved, helping to expand the game's universe by showing how the people from each of these worlds interact with each other. Some special story beats may also show why the tree was targeted in the first place.

Similar Solutions That Exist

2D Mario games play similarly, involving platform navigation and enemy encounters. In our game, players will enter levels that gradually increase in complexity.

Checkpoints will be strategically placed at key points, providing players with relief and a sense of accomplishment after overcoming challenging sections. This system ensures a smooth, engaging gameplay loop, allowing players to experience continuous satisfaction as they advance through obstacles.

Zelda games present puzzles and item unlocks that creatively expand the player's explorable world, creating a natural sense of progression and wonder. In our game, puzzles will serve as gateways that, once solved, allow players to access previously unreachable areas and new zones. Similarly, item unlocks will grant players additional abilities, opening up further exploration opportunities and deepening the sense of discovery as new paths and quest lines are revealed.

Hollow Knight has a similar blend of combat and platforming. It is not as technical and is more approachable by a casual audience. Our game will include features such as a three-hit grace window, allowing players to withstand up to three hits before experiencing consequences, and an optional assist mode to provide additional guidance during challenging sections. This ensures that players of varying skill levels can enjoy a satisfying combat experience. Hollow Knight feels very satisfying to accomplish platforming tasks.

Our game will draw on positive attributes from the games mentioned above and aim to incorporate them all to create an enjoyable experience in a less-explored niche of the gaming community. We intend to balance our game's difficulty to appeal to both casual players and those seeking a challenge. Our design will focus on approachability while also ensuring that gameplay remains satisfying through progressively more intricate challenges and a rewarding learning curve.

Feature List

Features That Will Be Completed

- Six fully playable levels
- Six unlockable items
- Various enemies
- Boss battle at the end
- Interactive platforming elements
- Saving and loading saved game file
- Healing
- Inventory space
- Inventory management system
- Checkpoint System

Features That May Be Completed If Time Allows

- Additional secrets in previous levels unlockable using items obtained later.
- Bestiary containing lore on all enemies, some obtainable items, and possibly some secrets
- Additional cutscenes showcasing other worlds not yet thrown into chaos (non-playable levels) when peering into their portals, with brief descriptions
- Personally create audio files like background and sound effects
- Optional assistance/guidance mode towards objectives
- Controller support vs. mouse and keyboard

Features We Want, But Can't Due to Time:

- Potentially additional levels, items, and enemies.
- Backstories for all NPCs.
- Downloadable via Steam platform, allowing cloud gaming and cloud sync for save files to be played across multiple computing systems.

Initial Set of Technologies

- Unity
  - Game development tool. The pot that mixes all the ingredients together.
- Windows OS
  - Choice of operating system for the game to run on.
- Github
  - Project team's method of collaboration during the development process.
- ProCreate
  - iPad application for artistic drawing which will be used for project design
- Audio Creation Software
  - e.i.: FL Studios, GarageBand, etc.
  - May be used for background music and sound effects if time allows.
- AI Tools
  - ChatGPT
  - Claude.ai
  - Google Gemini
    - Pro Licence Already Owned

Server Information

Malda will run as a standalone desktop application and not require any external services to function. Steam integration is a goal that we will not be able to accomplish due to time constraints but would require some of Steam's services for things like data syncing through their cloud platform.

Data Sources

Malda will not require data from external sources to be developed or function. Research on other products in the market may be done as a reference and development aide but none of this data will be required or directly used by the application.

AI Statement

Development Assistance

Malda will use AI tools like ChatGPT or Gemini to research and overcome obstacles in the development process. No code will be generated and directly used from AI sources.

Asset Generation

Malda will use Gemini to generate procedural audio and pixel artwork to be used directly in the game. Generated products may or may not be altered by hand before being utilized in the game.

Project Team Backgrounds

Eric Moore

- Programming since 2011 with a large variety of languages
- Unity experience with game development
  - Not since 2017
- Worked on multiple projects
  - Game Development
    - Java standalone
    - Unity
  - Web Development

- Collaborative experience with Github
- Minimal Pixel Art creation experience

Noah Turock

- Large experience in Java, some C experience (using a virtual machine), limited to no experience in others.
- Little to no experience developing a game or using Unity.
- Some sprite creation experience.

Rachel Mohamed

- Knowledge of a wide variety of programming languages and IDEs since 2022.
- Music and sound creation through FL Studios Mobile
- Knowledge/experience with sprite creation and game development through GameMaker Studios application and ProCreate.
- Little to no experience with Unity.

Anthony Chapman

- Knowledgeable in object-oriented programming languages and several IDEs and development tools
- Little experience with Unity, but some experience with C#
- Moderate experience with Git and Github
- No experience with sound, music, or asset creation

Dependencies, Limitations, Risks

Dependencies

- Core gameplay loop will need to be complete and polished before additional progress can be made
- Storyline is subject to change based around gameplay mechanics and development obstacles
- May require the use of audio libraries if time does not permit creation

Limitations

- Some levels or art/development concepts may be adjusted or removed/completely changed due to new project requirements or obstacles that may come up
- 2D design requires frame-by-frame animation instead of how 3D uses models.
- 2D development requires layering of backgrounds and foregrounds to create the illusion of 3D

Risks

- Limited prior knowledge of Unity
- Artwork and animation capabilities are low
- Group knowledge of GitHub is low and may result in challenges when collaborating

Concept Art

Below: Concept for Malda
Right: Concept for Tree Of Life